pfi

Python for Fixed Income

COURSE 2 · BOND MATH

The Price-Yield Relationship

WEEK 1 · DAY 3

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Above, at, and below 100

Day 2's four prices, each at its own yield, on 30 September 2026

Issuer	Coupon %	Yield %	Price
Pellwick Paper	4.75	4.6	100.4159
Quorvane Energy	5.25	5.4	99.3503
Dunmarrow Rail	6.125	6.125	100.0000
Halvering Foods	7.50	7.7	98.6227

The price is a function of the yield. When the yield moves, the price moves the other way.

Today: why a bond prices above, at, or below 100; how far each price moves; the pull to par; and a yield restated at another frequency.

Week 1: price and yield

Day 1	Time value of money
Day 2	Cash flows and price from yield
Day 3	**The price-yield relationship**
Day 4	Yield from price
Day 5	Case study: a new issue calendar

Par, premium, and discount

On a coupon date	Price	Here
coupon above yield	premium	Pellwick
coupon equal to yield	par, 100	Dunmarrow
coupon below yield	discount	Quorvane, Halvering

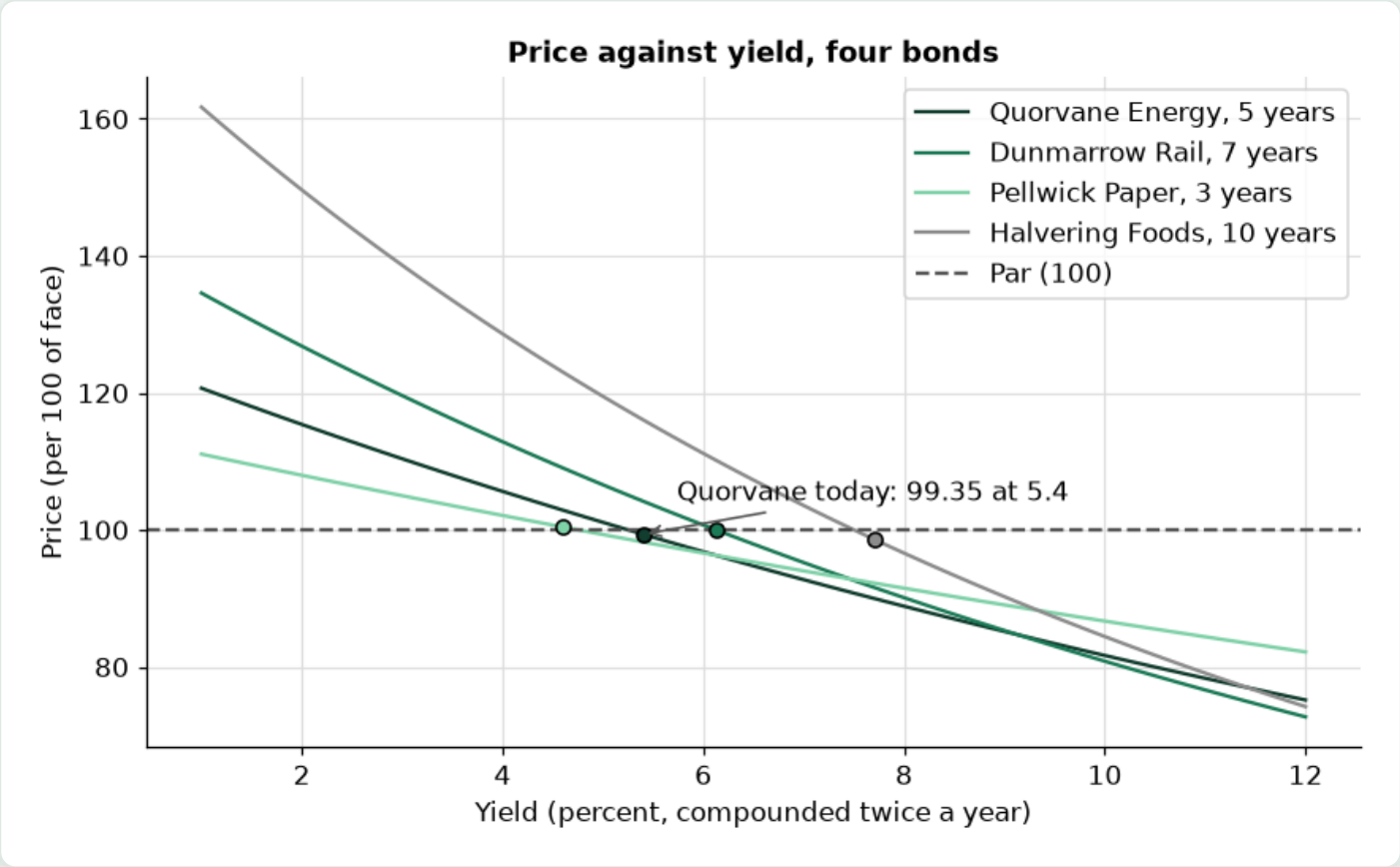
Coupon equal to yield: each coupon pays exactly the yield on the 100 lent.

A premium is the present value of the coupon's excess over the yield. A discount is the present value of its shortfall.



Pellwick: 0.075 per half-year, six times. Halvering: 0.100 short, twenty times.

The price-yield curve



+4.4276

Quorvane, yield 100 bp lower

-4.2055

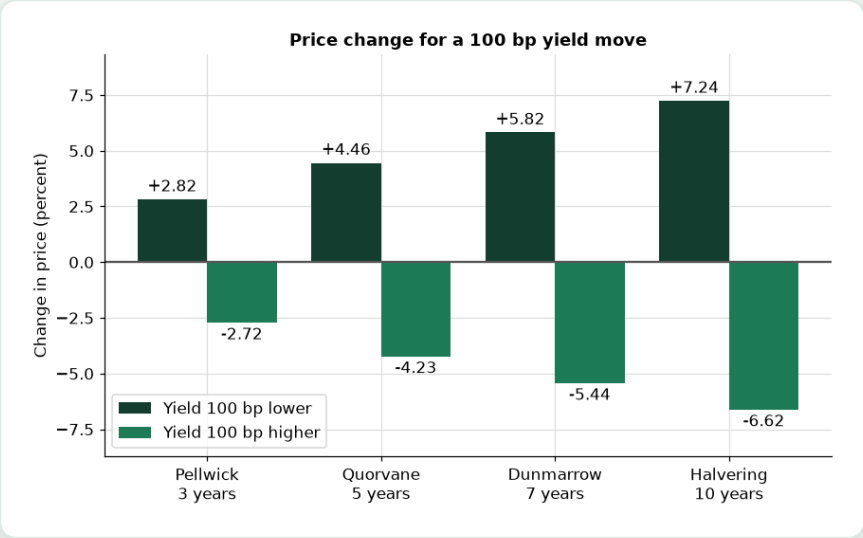
Quorvane, yield 100 bp higher

Price and yield move in opposite directions, and the curve bends.

The longest bond, Halvering, is the steepest; the shortest, Pellwick, the flattest.

Each line crosses 100 where the yield equals its coupon. The name for the bend is convexity.

Which bond moves more: yield up 100 bp



Maturity and coupon tables: what-ifs on Quorvane's terms, 5.25 percent coupon, 5.40 percent yield.

Maturity alone

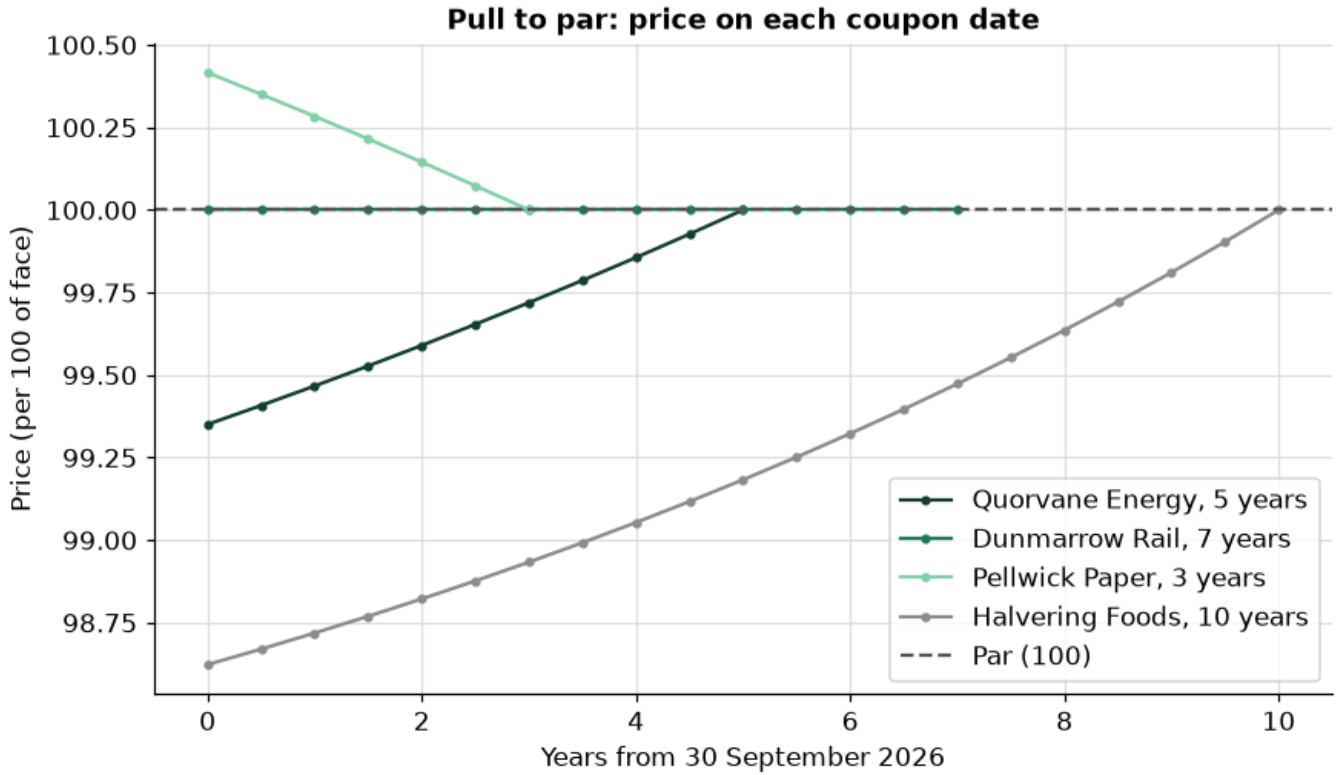
Years	Change %
1	-0.95
3	-2.70
5	-4.23
10	-7.34
20	-11.26
30	-13.33

Coupon alone, 5 years

Coupon %	Change %
0	-4.74
2.5	-4.47
5.25	-4.23
8	-4.05
10.5	-3.91

Longer maturity and lower coupon both push the weight of the cash flows later, so the price moves more.

Pull to par: the yield never moves



98.6227

Halvering today, at 7.70 percent

99.9037

Halvering with half a year left

Premium bonds drift down to 100, discount bonds drift up, a par bond stays at 100.

Halvering, next half-year: 3.85 percent, half of 7.70

If the yield does not change, the bond earns its yield.

Restating a yield: same growth in one year

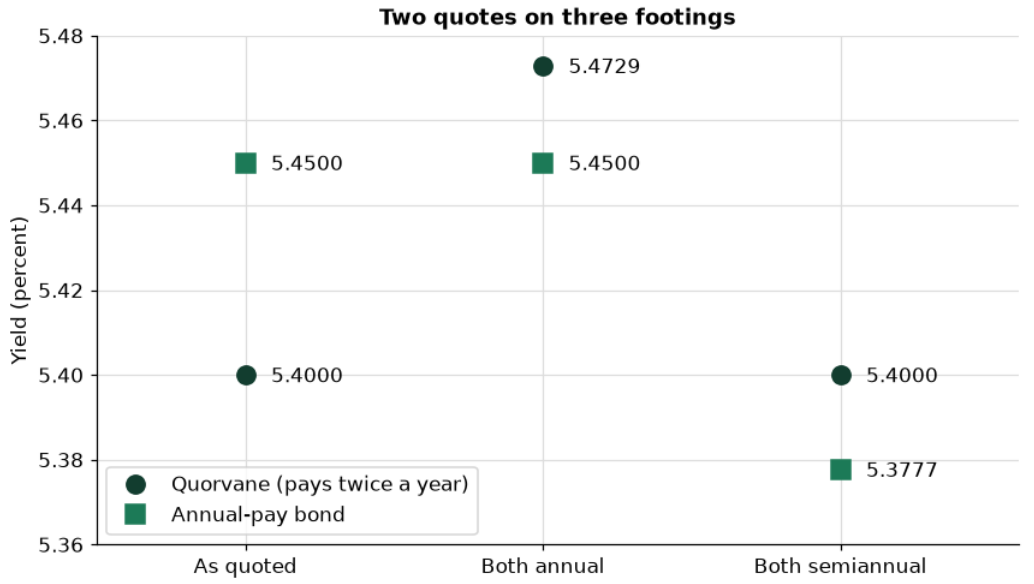
```
100 x (1 + 5.40/100/2)^2 = 105.4729
```

Compounded	Quorvane's yield %
once a year	5.4729
twice a year	5.4000
four times a year	5.3640

```
# convert_yield: one year's growth, then the new frequency
growth_one_year = (
    1 + yield_pct / 100 / from_frequency) ** from_frequency
period_rate = growth_one_year ** (1 / to_frequency) - 1
return period_rate * to_frequency * 100
```

Excel side by side (tested in Excel)

`=EFFECT(5.4/100,2)`	0.054729
`=NOMINAL(6.25/100,2)`	0.061553
`=NOMINAL(A1,4)*100`	5.364034, EFFECT in A1



Today's Excel, every formula tested

Job	Excel formula (tested)	Returns
Premium or discount	`=PRICE(DATE(2026,9,30),DATE(2029,9,30),4.75/100,4.6/100,100,2,0)`	100.415887
Price-yield column	`=PRICE(DATE(2026,9,30),DATE(2031,9,30),5.25/100,A2/100,100,2,0)`	113.35 to 87.35
The chart	`select A1:B8, Insert, Scatter with Straight Lines`	one line
100 bp up	`=PRICE(DATE(2026,9,30),DATE(2036,9,30),7.5/100,8.7/100,100,2,0)`	92.092797
Pull to par	`=-PV(7.7/100/2,2*A1,7.5/2,100), years left in A`	98.62 to 99.90
To annual	`=EFFECT(5.4/100,2)`	0.054729
To semiannual	`=NOMINAL(6.25/100,2)`	0.061553

Excel takes decimals: every formula writes out the /100, and a rate comes back as a decimal.

A message that says what and why

Part	Says	Today
Subject	what changed	Add convert_yield, tested against Excel EFFECT and NOMINAL
Body	why	yields at other frequencies must be restated to compare

```
git commit -m "<subject>" -m "<body>"
git log --oneline
git show --stat HEAD

    Add convert_yield, tested against Excel ...
 bondmath.py      | 19 +++++++
 test_bondmath.py | 18 ++++++
```

Write the subject as an instruction: it completes the sentence "This commit will ...". git log --oneline shows only the subject.

In six months, git log is how you find the change that matters.

Bring it home: paste, test, read, commit

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Today's three append cells at the end of your two files, without the %%writefile line.

3

Test

python -m pytest -q: 8 passed, or 10 with the exercises of days 1 and 2.

4

Commit

git diff --stat, git add, then git commit with a subject and a body.

5

Read

git log --oneline and git show --stat HEAD. Press q to leave a pager.

Pasting at the end keeps anything of your own. Copy fallen behind? Take the start-of-day files from today's setup cell, and git diff shows what changed.

In Colab: download bondmath.py and test_bondmath.py from the work folder again. They replace yesterday's downloads.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert against Excel's reference rows. Run it: it fails.

4

Fix

Find the convention it broke, fix it, run the test again.

Today: convert_yield uses the percent figure as a decimal, and the round-trip test still passes.

Code that runs is not the same as code that is right.

What you can do now

- Say from coupon and yield whether a bond is at a premium, par, or a discount
- Draw the price-yield curve in matplotlib and Excel, and say why it bends
- Say which bond moves more: the longer one, and the lower coupon
- Show the pull to par, and that a bond earns its yield if its yield holds
- Restate a yield with convert_yield, EFFECT, and NOMINAL; commit what and why

Week 1, Day 4 Yield from price: Goal Seek, a bracket halved by hand, and a loop